

Compressed UQFF with B/B_{crit} Superconductivity

Daniel T. Murphy

Independent Research • Star-Magic / UQFF Project

Session: 101 • Source: grok_share_11254865.txt (lines 2700-3400) • UQFF v4.74

Abstract

This paper presents the Compressed UQFF formulation incorporating Newtonian gravity, Hubble expansion, superconductivity via the B/B_{crit} flux-quenching factor, environmental coupling, cosmological constant contribution, quantum coherence, fluid dynamics, and dark matter perturbation. Parameterised for seven astrophysical systems and validated against SGR1745-2900. This is the first UQFF formulation to explicitly incorporate Bardeen-Cooper-Schrieffer (BCS) superconductivity quenching into the gravitational coupling (Linear Meissner form; see PAPER_375 for the exponential form).

1. Master Equation

The Compressed UQFF master gravity equation combines eight modular terms:

$$\begin{aligned} g_{\text{UQFF}}(r, t) = & GM(t)/r^2 * [1 + H(t, z)] * [1 - B/B_{\text{crit}}] * [1 + F_{\text{env}}] \\ & + (Ug1 + Ug2 + Ug3' + Ug4) + \Lambda c^2/3 \\ & + (\hbar/dx dp) * \text{integral}(\psi * \hat{H} \psi dV) * 2\pi/t_{\text{Hubble}} \\ & + \rho_{\text{fluid}} * V * g + (M_{\text{vis}} + M_{\text{DM}}) * (d\rho/\rho + 3GM/r^3) \end{aligned}$$

where $H(t, z) = H_0 t$, $H_0 = 2.269 \times 10^{-18} \text{ s}^{-1}$ (Newtonian cosmological expansion approximation).

2. Modular Component Functions

Function	Formula	Key Constants
compressed_base	$G * M / r^2$	$G = 6.674 \times 10^{-11}$
compressed_expansion	$1 + H_0 * t$	$H_0 = 2.269 \times 10^{-18} \text{ s}^{-1}$
compressed_super_adj	$1 - B/B_{\text{crit}}$	Linear Meissner form
compressed_env	1.0	Default (no environment coupling)
compressed_cosm	$\Lambda * c^2 / 3$	$\Lambda = 1.1 \times 10^{-52} \text{ m}^{-2}$
compressed_quantum	$(\hbar / 1e-68) * 2.176 \times 10^{-18} * (2\pi / t_H)$	$t_H = 4.35 \times 10^{17} \text{ s}$
compressed_fluid	$\rho_f * V * g_{\text{local}}$	from MUGESystem
compressed_perturbation	$(M + M_{\text{DM}}) * (d\rho/\rho + 3GM/r^3)$	$d\rho/\rho = 1e-5$

Table 1. Compressed UQFF modular component functions

3. Seven Astrophysical System Parameters

System	M (kg)	r (m)	B (T)	B_crit (T)	V_sys (m3)
SGR1745-2900	2.984e30	1e4	1e10	1e11	4.189e12
Sagittarius A*	8.155e36	1e12	1e-5	1e-4	3.552e45
Tapestry Starbirth	1.989e35	3.086e17	1e-4	1e-3	1e53
Westerlund 2	1.989e35	3.086e17	1e-4	1e-3	1e53
Pillars of Creation	1.989e32	9.46e15	1e-4	1e-3	3.552e48
Rings of Relativity	1.989e36	3.086e17	1e-5	1e-4	1e54
Student's Guide Universe	1e53	1e26	1e-10	1e-9	1e80

Table 2. MUGESystem parameters for seven astrophysical systems

4. Validation

Unit test: test_compute_compressed_MUGE(SGR1745-2900)

Expected: $1.782 \times 10^{39} \text{ m/s}^2$

Dominated by compressed_base $\times$ expansion; $B/B_{\text{crit}} = 0.1 \rightarrow 90\%$ retention under linear Meissner form.

5. Physical Interpretation

The $[1 - B/B_{\text{crit}}]$ factor models the Meissner effect: as the magnetic field B approaches the critical field B_{crit} , gravitational coupling is quenched — mirroring how a Type-I superconductor expels magnetic flux below B_{crit} . The Compressed UQFF thus unifies gravitomagnetic quenching with cosmological expansion and quantum corrections in a single framework.

For Type-II exponential treatment (London penetration depth model) see PAPER_375.

6. Implementation

C++: STAR_MAGIC_09SEPT_UQFF_MODULE.cpp, namespace CompressedUQFF

Python: CondensedPhysics4.py, class CompressedUQFFBcritSuperconductivityCalculator (CP4 #20)

Wolfram macro: WOLFRAM_TERM_COMPRESSED_UQFF_BCRIT